

Program 25: Plot Multiple Lines

AIM

To write a Python program to plot two or more lines with legends, different widths, and colors.

CODE

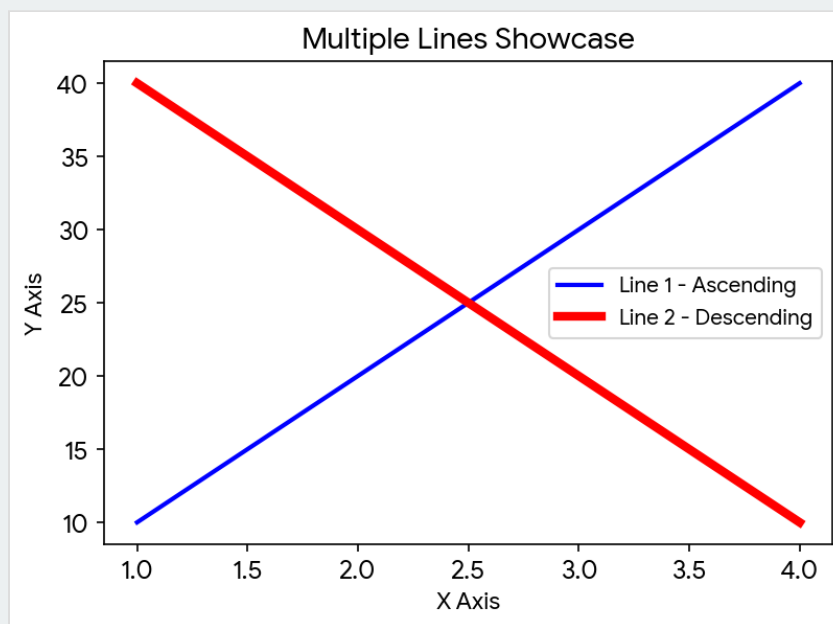
```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4]
y1 = [10, 20, 30, 40]
y2 = [40, 30, 20, 10]

plt.plot(x, y1, color='blue', linewidth=2, label='Line 1 - Ascending')
plt.plot(x, y2, color='red', linewidth=4, label='Line 2 - Descending')

plt.xlabel('X Axis')
plt.ylabel('Y Axis')
plt.title('Multiple Lines Showcase')
plt.legend()
plt.show()
```

OUTPUT



Result: Multiple lines with specific formatting and a legend were successfully plotted on the same axes.

Program 26: Create Multiple Plots

AIM

To write a Python program to create multiple subplots within a single figure.

CODE

```
import matplotlib.pyplot as plt

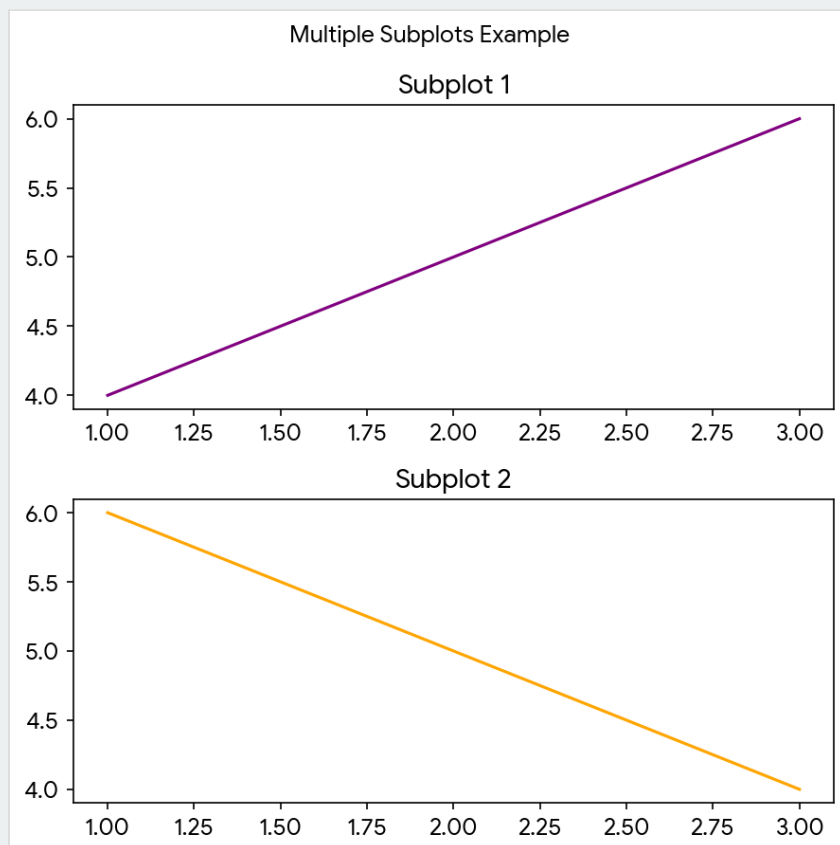
fig, axs = plt.subplots(2, 1)
fig.suptitle('Multiple Subplots Example')

axs[0].plot([1, 2, 3], [4, 5, 6], color='purple')
axs[0].set_title('Subplot 1')

axs[1].plot([1, 2, 3], [6, 5, 4], color='orange')
axs[1].set_title('Subplot 2')

plt.tight_layout()
plt.show()
```

OUTPUT



Result: A multi-plot figure was successfully created and cleanly formatted using tight_layout.